

An overview of our chamber problem

Broadband directivity works from 125 Hz up; below 2 kHz the room bends the blade tones by up to 8 dB. For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height in the propeller plane, so a perfect setup reads the same level everywhere: a circle. Twelve runs (prop7-18, 1-2 Sept 2026, PWM 1900); what is left after each microphone's own offset is removed is the room.

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2026-09-08 | rev 2
data: laptop backup of 2 Sept
scripts: arc_error_map.py, arc_validation_diagnostics.py
         full guide: docs/arc-validation-guide.html
remedies report: docs/arc-validation-remedies.pdf

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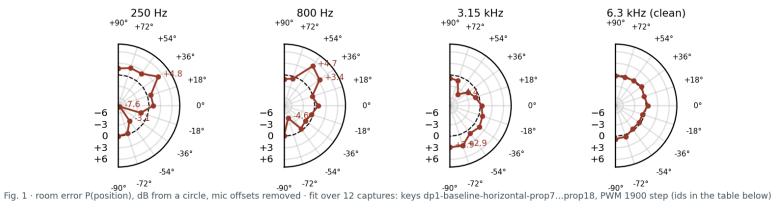


Fig. 1 · room error P(position), dB from a circle, mic offsets removed · fit over 12 captures: keys dp1-baseline-horizontal-prop7...prop18, PWM 1900 step (ids in the table below)

What the room does (right panel of Fig. 2)

How to read the map. Row = spot on the arc, column = frequency band. White: the spot reads what a circle predicts. Red: too loud. Blue: too quiet. Each microphone's own error is already taken out (left panel), so the colour is the room

Deep blue at -72° and red at $+36^\circ/+54^\circ$, 500-1000 Hz. A sound wave that bounces off a nearby surface arrives a little later than the direct wave. If the extra path is half a wavelength the two cancel and the spot goes blue; if it is a whole wavelength they add and the spot goes red. The extra path depends on the spot, so neighbours can be opposite colours and one spot can flip colour from band to band. Wavelength here is 35-70 cm, so the surface is within about a metre.

The 200-250 Hz column is the blade tone, and it is not one nearby surface. At BPF 216 Hz the tone polar is round to ± 2 dB; at 238 Hz -36° reads -8 dB; at 258 Hz the hule moves to -72° and the peak to $+54^\circ$, while the broadband floor beside the tone stays round to ± 1 dB. A surface within a metre cannot change the pattern that fast; narrow room modes sampled by a line spectrum can, and so can a propeller whose inflow is not axisymmetric. A loudspeaker at the hub decides which (remedies report, \$03, \$10).

Blue on one side, red on the other: the surface is off-centre. A surface symmetric around the propeller (the floor under a flat arc, say) gives every microphone the same extra path, so a whole row would be one colour. Here the colour changes around the arc, so the microphones are at different distances from the surface: it sits to one side. The colours do not say which side; blue is not "near" and red is not "far". Proceeding with the tone at three speeds: the $+36^\circ/+54^\circ$ reflection has an extra path of 1.5 m and a hard surface's amplitude, the -72° one 0.7 m (remedies report, Fig. 3); the unrecorded arc radius is what turns those into places.

White above 4 kHz and white everywhere once the tones are taken out. A 900 Hz-wide band at 4 kHz averages a reflection out; a 58 Hz band at 250 Hz holds one cancellation, so a tone shows it. With every tone notched, the broadband polar is round to ± 1.3 dB in 92 % of cells from 125 Hz up. **This is not a conclusion to discard everything below 4 kHz:** broadband directivity is usable from 125 Hz now; only blade-tone levels below ~ 1 kHz need the loudspeaker check, the speed ladder or the measured correction first (remedies report \$00, \$03).

Fig. 2 · Microphone error map (left) and room error map (right), same fit, same scale

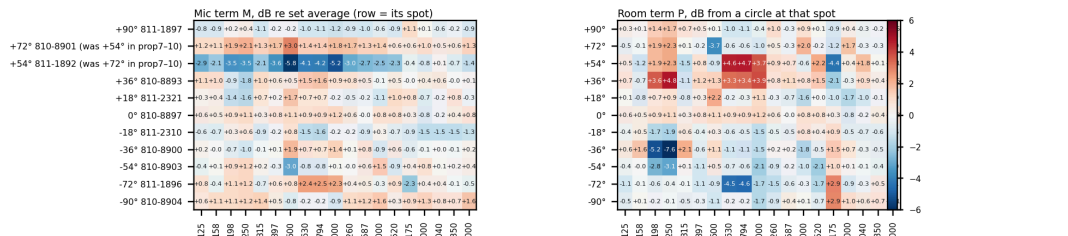


Fig. 2 - one fit over keys dp1-baseline-horizontal-prop7...prop18, PWM 1900, prop11-14 mirrored, rows aligned: mic row i sat at spot row i - 0° mic never moved, so its M and the 0° P are known only as a sum - third-octave centre, Hz

What to look for in the chamber

Three suspects, one per band in Fig. 2. Each is a size and a place, read off the map: a surface shows up once the wavelength is shorter than it (Rajmane and Baumann, DAGA 2016, Table 1: a 35 cm plate from 1 kHz, a 10 cm plate from 3.15 kHz).

1. **200-250 Hz, blue at -36°.** Something about 1.5 m across, off to one side of the rig, with a bounced path 0.7 m longer than the direct one at -36°: a wall, a table top or the corner within about half a metre of that microphone.
2. **630-1000 Hz, blue at -72°.** Something about 0.4 m across near the -90° end of the arc on the corner side: the stand base, a bar of the arc frame, a cable tray.
3. **3150 Hz, scattered red and blue.** Something about 0.1 m across at the microphones themselves: clamps, mounts, the cable loop.

Next test (one afternoon)

1. Take 811-1892 off the arc.
2. Put a small loudspeaker on the stand at the hub. Play a 20 Hz–20 kHz sine sweep and record all microphones; each microphone's own impulse response gives its echo delays (no clock sync needed), times 343 m/s, the extra paths. Then play tones at 216, 238 and 258 Hz: if the ± 8 dB pattern of Fig. 1 comes back, it is the room; if not, it is the propeller's inflow.
3. Write down the geometry in docs/mic_arc.md: arc radius, propeller height, which wall or corner is at the -90° end, distances from the -72° and $+72^\circ$ microphones to the nearest wall.
4. Run a ladder of 12–15 motor speeds (BPF every ~ 5 Hz, 205–265 Hz) and average the tone polar over it; cover or move the surfaces the sweep names; rerun the fit. Foam works from 650 Hz up; at 250 Hz the levers are averaging, correction and distance, then resonant absorbers.

How the numbers were made (not a sum, not an average)

Every level in the twelve runs (11 microphones $\times$ 19 third-octave bands $\times$ 12 runs) is written as three parts added together: **run gain** + **position error** + **microphone offset**. The three parts are found by least squares over all 12×11 levels in every band. **Run gain** takes up any level difference between runs (RPM, day, load). **Microphone offset** takes up each serial's own error after calibration. **Position error** is what is left that belongs to a spot on the arc; it is forced to average zero over the eleven spots, so it reads as dB from a circle. That is what Figs. 1 and 2 show.

The turned-around runs are handled before thfit: for prop11-14 the label +e is entered as physical -e, so "-72°" always means the same spot in the room. The four microphone offsets and the eleven position errors can be told apart only because the same spot was covered by different microphones (the swap after prop10 and the four turned runs). Without that, a low microphone and a low spot would be indistinguishable. Spectra: Welch 4096-point Hann, UMIK-2 calibration as the app applies it, third-octave sums 125 Hz-8 kHz. Data: the last PWM-1900 capture in each key 2004_6in_unset_dpi-baseline-horizontal-prop7-prop18, 11 mics x 2.0 s; prop11-14 with the arc turned around; prop1-6 and the 31 Aug keys left out (they match neither orientation). Capture ids per key: docs/analysis/arc-validation-data.json. Code: scripts/arc_error_map.py, function fit.

Do the numbers add up?

Refitting from the data pack reproduces the maps to 0.01 dB; each map sums to zero per band by construction. 132 cells per band are explained by 34 unknowns; 87% of them are within 2 dB of gain + P + M. Scatter is 0.6–0.8 dB in the quiet bands and 1.5–2.3 dB in 200–250 Hz and 500–1000 Hz, the bands where the room acts. The worst cells (+11 dB at $-36^\circ/250$ Hz in prop7; +6 to +7 dB at $0^\circ/2\text{--}4$ kHz in prop7–8) come from prop7, which spun at BPF 164 Hz where every other run spun at 238 Hz: a different speed; leaving it out moves the maps by under 1 dB. Limit: the 0° microphone (810-8897) never moved, so its M and the 0° P are known only as a sum.

Worked example · -72° spot at 800 Hz, two mics, two orientations

RUN	MIC THERE	MEASURED	GAIN	P	M	MODEL	LEFT
prop18	811-1896	59.1	62.3	-4.6	+2.5	60.2	-1.1
prop12 (180°)	810-8901	58.6	62.2	-4.6	+1.4	59.0	-0.4
+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4; prop18 with 811-1892 reads +3.1 = +4.7 - 4.2. Two mics, one room value.							